

Head-up Displays for Augmented Reality Applications in Racing Vehicles: a Feasibility Study

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Abstract— This paper studies the applicability of Augmented Reality Head-Up Displays (HUD) to car racing applications. HUD have been used for years in avionics to effectively represent environmental cues to the pilot without the need to look at the instruments. Recent advances in technology claim that similar systems can be developed for cars. HUD have been available for cars for some time, but have been exploited only as virtual dashboards. This paper looks into the potential benefits and limitations of truly Augmented Reality for car racing. In particular, the objective is to highlight the curbs of the track so to help the driver plan the best trajectory. We develop the algorithms needed to detect the curb and project its position on the HUD. We show that, in order to provide a precise projection of the cue, the attitude of the vehicle needs to be estimated and the information from a camera needs to be merged with a map of the racing track. We test the approach on an experimental vehicle rigged to emulate a HUD. The results indicate that, while the algorithms can be designed to be robust and precise, they require a very large Field of View.

I. INTRODUCTION

The few past years have marked the commercial success of Advanced Driving Assistance Systems (ADAS). ADAS improve safety, efficiency and pleasure of driving. ADAS can be classified as either active or passive. Active ADAS have actuators that change the state of the vehicle. Automatic Braking, Lane Keeping and Adaptive cruise control are all examples of active ADAS. Passive ADAS, on the other hand, help the driver better assess the situation without affecting the vehicle state. Examples are: parking cameras, lane departure warning systems, blind spot warning. Most passive systems use haptic [1], sound or blinking lights to warn the driver. These sensory channels are suitable to communicate simple warnings. They cannot convey complex situational information. *Augmented Reality* (AR) could overcome these limitations. Augmented reality, whose development is driven mainly by industrial or gaming applications, consists in the idea of overlaying synthetic information on a view of the real world [2]. The application of AR to vehicles dates back to 1942 [3] even before it was termed AR: the British night fighter de Havilland Mosquito was provided with a Head-Up Display (HUD), a transparent screen over which some information is projected. Since then, significant effort has been invested to design both military and civil applications in avionics [4], [5].

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From aviation, the use of HUD's has interested other vehicles. GM introduced the first automotive HUD as an option on the 1988 Oldsmobile Cutlass Supreme. HUD's have been included in the standard configuration of more recent vehicles like the Peugeot 3008 and Chevrolet Camaro. These devices inform the driver about, for instance, speed and available fuel; they could also provide some basic navigation cues. They aim at enhancing safety by keeping the driver's gaze on the road. However, as pointed out in [6], these pictograms induce the so-called *tunneling effect*, leading to poorer driving performance. These HUD's are not applications of AR; the information is shown without any relation to the real world, see Fig. 1. They simply replicate part of the dashboard at the driver's eye level.

More recently, WayRay has introduced the concept of HUD 2.0 on the market. This product is closer to the idea of AR [7]. The Field of View of the HUD is now sufficiently wide to overlay information on the real view. Figure 1 shows a proof-of-concept of a representation of the best trajectory for racing applications. This technology extends the range of possible applications: from seeing through the fog to suggesting a safety manoeuvre to avoid an accident or other road hazards [8].

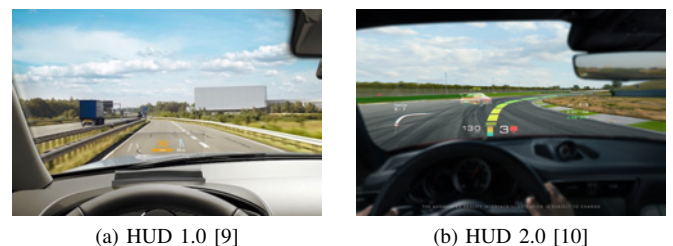


Fig. 1: Overview of the two approaches to HUD currently available on the market.

Some research groups have investigated the potentials of HUD 2.0 (or contact analog head-up displays - cHUD - as they are sometimes referred to) in laboratory settings even before a viable technology was available. The most relevant contributions are [11]–[15]. In [11], the authors design an advanced navigation framework to be projected on the HUD, improving the effectiveness of the standard indicators. In [12], the authors investigate an AR framework which provides two synthetic features describing the forecasted motion of the vehicle. The results of [13], using a simulated environment, prove that the precision with which the AR element is seen in relation to the real environment is critical

for the effectiveness of the cue. Rusch and co-authors [14] address the concern that the AR cues could distract the driver from other non-cued information. Their data does not support that this is a risk. The literature agrees on the potentials of HUD 2.0 and strongly correlates the effectiveness of the system with its accuracy. It is interesting to observe that all the cited contributions focus on city driving and most use simulated data.

This paper pushes the field of application of the HUD technology by investigating racing assistance systems. The main contribution is the development of a proof-of-concept useful to analyze the required computational methods and understanding the potentials and limitations of the technology based on real data collected on a racing track. In more detail, we aim to highlight the circuit curbs so to help the driver plan the best trajectory for the corner. When racing, making the apex of the ideal racing line is important for performance. Having a clear indication of the curbs would help drivers improve their trajectories. In this work, we will emulate the effect of the system by projecting the cues directly on the view of a camera placed where the driver's head would rest; in this way it is possible to better assess the requirements of the HUD hardware. Note that for safety reasons, but without loss of generality, we will actually consider the passenger's point of view.

The outline of the paper is as follows: Section II defines the goals of the study and the experimental setup. Section III illustrates the methods exploited in the current application and the algorithms designed to make the feature detection possible. Experimental results are discussed in Section IV, followed by some final remarks.

II. PROBLEM STATEMENT AND EXPERIMENTAL SETUP

This feasibility study is based on data collected during experiments performed with a sports car on the closed racing track of Castelletto di Branduzzo. The vehicle, an Alfa Romeo Giulia Veloce, is equipped as follows:

- a Suchy System X Pro GPS nano, which measures the vehicle position at 10 Hz;
- a 6-degrees-of-freedom Inertial Measurement Unit (IMU) sampled at 100 Hz;
- the On-Board Diagnostic (OBD) CAN interface, which records the vehicle speed;
- a standard Advanced Driver-Assistance System (ADAS) camera, placed along the vehicle centerline under the rearview mirror, rigidly connected to the vehicle;
- an additional camera, replicating the driver's perspective, rigidly connected to the seat. This camera is used to represent and visualize the system capabilities.

All sensors are logged by means of a NVIDIA DRIVE board, in the configuration illustrated in Fig. 2.

It is assumed that the IMU is mounted perfectly aligned with respect to the standard reference system (Fig. 2) and any mismatch is compensated following the algorithm proposed in [16].

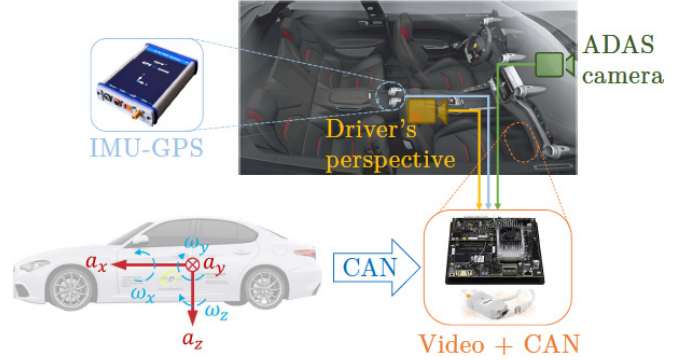


Fig. 2: Schematic of the experimental setup and the vehicle's standard reference frame.

III. METHODOLOGY

The implementation of the curb highlighting system requires two steps: first, we explain how to project onto the HUD a feature whose spatial coordinates are known so that the driver would see the object in the correct place in their field of view. Secondly, we describe the curb detection algorithm that computes the world coordinates of the feature.

A. Feature Projection

In the following discussion, we will refer to the vehicle-centered coordinate system, the ADAS camera reference frame and the HUD reference frame. As the name implies, the vehicle-centered coordinate frame's center is at the vehicle center of mass; its z axis rotates with the vehicle so that its x axis is always directed as the vehicle direction and the other two axes are parallel to the road surface in that position. In other words, the vehicle-centered coordinate system follows the position of the vehicle and its direction, but it is insensitive to the vehicle's pitch and roll. The coordinates of an object in this reference frame are $\mathbf{w} = [x_w \ y_w \ z_w]^T$.

The ADAS reference frame is rigidly connected to the ADAS camera which is in turn rigidly connected to the vehicle's frame. Assuming a pin-hole model [17] of the camera, any point with known vehicle-centered coordinates $\mathbf{w}$ is projected onto the ADAS camera's image plane at coordinates $\mathbf{a} = [u_a \ v_a]^T$ according to

$$\begin{bmatrix} u_a \\ v_a \\ 1 \end{bmatrix} = K_a \begin{bmatrix} R_a(\varphi, \theta) & \mathbf{t}_a \end{bmatrix} \begin{bmatrix} \mathbf{w} \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} f_{a_x} & s_a & c_{a_x} \\ 0 & f_{a_y} & c_{a_y} \\ 0 & 0 & 1 \end{bmatrix}}_{\text{intrinsic}} \underbrace{\begin{bmatrix} r_{a11} & r_{a12} & r_{a13} & t_{a_x} \\ r_{a21} & r_{a22} & r_{a23} & t_{a_y} \\ r_{a31} & r_{a32} & r_{a33} & t_{a_z} \end{bmatrix}}_{\text{extrinsic}} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}, \quad (1)$$

in which the two matrices are termed *intrinsic* and *extrinsic* parameters matrix, respectively. The intrinsic parameters contain the information of the focal length (f_{a_x} , f_{a_y}), the optical center (c_{a_x} , c_{a_y}), and the skew coefficient (s_a). The extrinsic parameters describe the relative attitude and

location of the camera reference frame with respect to the vehicle-centered frame. Based on the above assumptions, the translation part of the extrinsic parameter is fixed and can be easily calibrated once for all [18]; the attitude part depends on the roll, pitch and yaw angles of the ADAS reference frame with respect to the vehicle-centered frame: the yaw angle is affected only by the mounting angle and, as the translation, is calibrated. The roll (φ) and pitch (θ) angles are affected by the chassis movements (which, especially in racing applications, cannot be neglected) and will be estimated online.

Thanks to (1), it is possible to map features expressed with a set of coordinates in three dimensions into their two-dimensional pixel coordinates counterparts, meaning that we can relate the geospatial knowledge with the camera information, a crucial process for the feature detection algorithm discussed later on.

The third reference frame is the HUD reference frame. HUD's are characterized by a head motion box (HMB) [2]. The HMB defines a region of space such that if the observer's eyes lie in this region, the shown cue is not affected by parallax: it will overlay the same world position regardless of the actual position of the observer. The procedure of boresighting [19] returns the transformation needed to correctly project points in the HUD. In this work, we emulate this parallax-free effect by rigidly mounting the driver's perspective camera to the car and by defining the HUD frame as a frame centered on the driver's perspective camera. In this way, the world coordinates expressed with vector $\mathbf{w}$ are mapped into the HUD coordinates vector $\mathbf{d} = [x_d \ y_d \ z_d]'$ through

$$\mathbf{d} = [R_o(\varphi, \theta) \quad \mathbf{t}_o] \begin{bmatrix} \mathbf{w} \\ 1 \end{bmatrix} \quad (2)$$

$$\begin{bmatrix} x_d \\ y_d \\ z_d \end{bmatrix} = \begin{bmatrix} r_{o11} & r_{o12} & r_{o13} & t_{ox} \\ r_{o21} & r_{o22} & r_{o23} & t_{oy} \\ r_{o31} & r_{o32} & r_{o33} & t_{oz} \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix},$$

in which $\mathbf{t}_o \in \mathbb{R}^{3 \times 1}$ is the translation vector, setting the reference system at the center of the driver's perspective, and $R_o \in \mathbb{R}^{3 \times 3}$ is the rotational matrix, which aligns the vehicle-centered frame with respect to the driver's one. It is worth remarking that in this case, we assume that the driver's point of view is rigidly connected to the ADAS camera point of view and its relative position is apriori known. Of course this is in general not true, but in the real scenario these hypotheses are not needed thanks to the HUD optics which guarantee a parallax-free projection. An overview of the two reference systems and the roto-translation is illustrated in Fig. 3.

In our framework, we emulate the HUD by representing information directly on the driver's camera image plane. By modeling this driver's camera as a pinhole camera, features whose HUD frame coordinates are known can be mapped

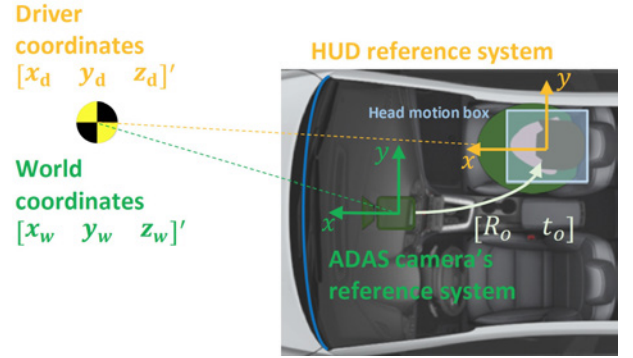


Fig. 3: Representation of the reference systems.

onto their corresponding pixels as

$$\begin{bmatrix} u_d \\ v_d \\ 1 \end{bmatrix} = K_d \mathbf{h} = \begin{bmatrix} f_{d_x} & s_d & c_{d_x} \\ 0 & f_{d_y} & c_{d_y} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_h \\ y_h \\ z_h \end{bmatrix}. \quad (3)$$

This operation is performed only for the sake of representing the results, as the person would be in place of the camera in the real context. Figure 4 represents a simple example of

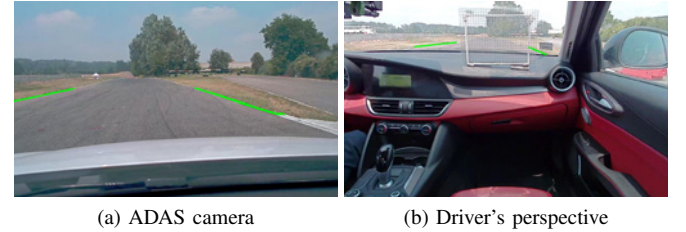


Fig. 4: Visual validation of the HUD feature detection and projection process.

the proposed projection method. The figure represents the results of a multi-step process:

- 1) A standard lane detection algorithm [20] is applied to the image captured by the ADAS camera. This returns the pixel coordinates of the lanes.
- 2) The pinhole model (1) is inverted assuming that the lane are on the ground and that the vehicle is not affected by neither roll nor pitch. This returns the vehicle-centered coordinates of each point belonging to a lane.
- 3) Each lane's detected point is roto-translated and projected according to Equations (2 – 3).
- 4) The results are shown in the driver's camera.

As it can be seen from the figure, the reprojected lines perfectly overlaps the road lanes in the driver's camera. Note that, in this process, the data stream of the driver's camera is not used. The image is only used to visualize the results. In the example, the computation of the vehicle-centered coordinates of the feature and the application of the inverse perspective mapping are greatly simplified. In the remainder of the paper, these issues will be studied in more details.

B. Curb detection

The primary goal of this proof of concept is to highlight the racing track curbs; this section discusses the details of the detection process.

1) *Inverse perspective mapping and attitude compensation*: The first phase of the detection algorithm removes the perspective effect of the ADAS camera. This is possible by applying the geometrical transformation known as Inverse Perspective Mapping (IPM) [21], which removes the perspective effect from the acquired image, remapping it into a 2-dimensional domain, termed Bird's Eye View (BEV), in which the information content is homogeneously distributed among all pixels.

Our method is based on the definition of a Region of Interest (ROI) and on the assumption that there is no variation of road slope and inclination in this ROI. Given these hypotheses, the world coordinates of objects that appear in the ROI that lie on the road level can be estimated through the BEV.

However, as pointed out in Section III, in order to apply the IPM (*i.e.* inverting (1)) one needs to know the vehicle roll and pitch angles (see Fig. 5). This effect cannot be considered negligible in motorsport as the vehicle experiences significant longitudinal and lateral dynamic load transfer. Several

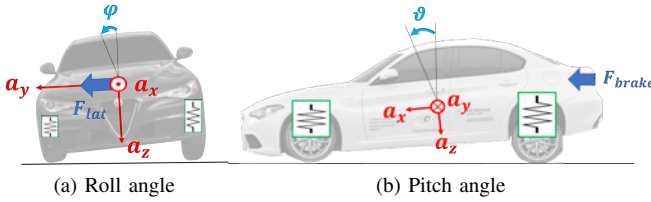


Fig. 5: Effects of the driving dynamics on the vehicle's attitude.

methods to estimate the roll and pitch angle exist, one of the simplest is a static estimation based on the so called roll and pitch stiffness of the vehicle. Following the method proposed in [22], we estimate the roll and pitch angles according to:

$$\varphi \approx -\frac{Mh}{K^s L_\varphi^2} a_y = \gamma_\varphi a_y \quad (4a)$$

$$\theta \approx -\frac{Mh}{K^s L_\theta^2} a_x = \gamma_\theta a_x \quad (4b)$$

in which M is the vehicle mass, h is the center of mass height, K^s the suspension stiffness, L_φ and L_θ the vehicle's track and wheelbase, respectively. These parameters are specific for the given car and, in this study, are provided by the car maker. Figure 6 plots the reconstructed roll and pitch angle during a portion of the track. The figure shows that roll angles up to 3.5 degrees are reached.

Once the roll and pitch angles are known, they can be incorporated in the IPM. Figure 7 exemplifies the importance of this estimation: it shows the BEV obtained without accounting for the roll and pitch angle (on the right), and the

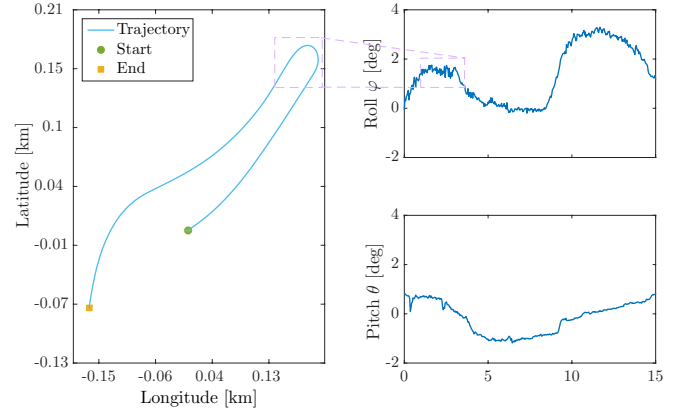


Fig. 6: Analysis of a portion of a lap in which the roll angle during a turn is highlighted.

one with the compensation. It is clear that the uncompensated BEV is skewed and imprecise.

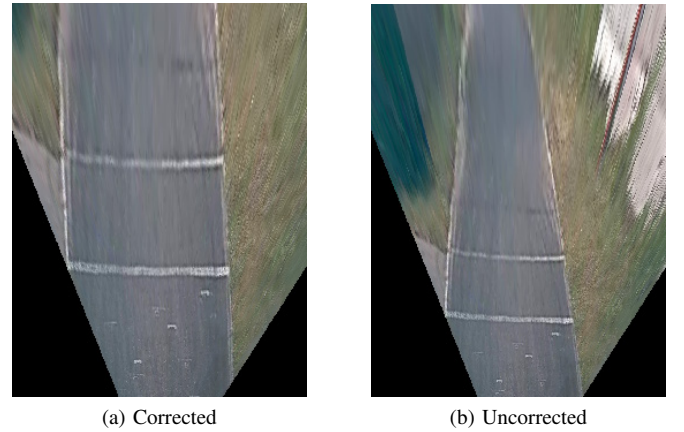


Fig. 7: Comparison of the BEV computed with the attitude correction vs the uncorrected one.

2) *Feature detection*: Figure 7 also shows that a detection of the curbs based only on the ADAS camera would not be useful. Because of the low height of the vehicle, the ROI extends at most to 60 meters ahead of the vehicle; moreover, the farther from the camera, the less accurate the IPM is. The driver needs a longer preview to set a good trajectory. To solve this issue, we propose a hybrid GPS/GNSS - computer vision based curb detection algorithm. The idea is to compute the coordinates of curb in the vehicle-centered reference frame using map-based methods when the vehicle is far from the corner; and using vision-based methods when the curbs is in the ROI.

Racing tracks are usually well mapped; it is thus possible to select the GPS coordinates of the points belonging to the curb. A roto-translation that accounts for the vehicle GPS and its heading returns the coordinate of the feature $\mathbf{w}_{pos}$ where the pos subscript indicates that the coordinates are obtained through the positioning system.

As the vehicle gets closer to the corner, the geolocation

errors would show up in the projection as misalignments. This issue can be solved by detecting the curb using computer vision techniques when the curb is in the ROI. If the curb is a wide and dashed line (as it is on all circuits), it can be detected following the approach proposed in [20]: given the curb width, the algorithm segments the BEV image into objects with similar dimensions, highlighting the only ones corresponding to the target size. In the example of Fig. 8b, this process provides the green spots.

Although effective, this visual segmentation comes with some noise. To remove this effect, points are discretized into parallel lines equally spaced in one dimension, while the other dimension is unconstrained. The discretization step obtains the edges of the curb's alternating-colored dashes, whose length correspond to the width of the squares, known for a given circuit. The results of this operation is remarked with the blue lines in Fig. 8b: as shown, this generalization step provides a synthetic and less noisy detection of the curb.

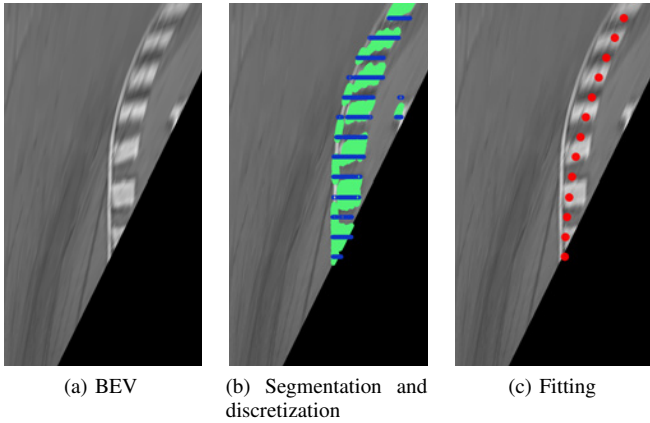


Fig. 8: Step-by-step illustration of the curb detection from the camera image.

Finally, the profile of the curb is obtained by fitting a second-order polynomial on the extracted features in the BEV coordinates (*i.e.*, the discretized lines) thanks to their points distribution. The outcome of this operation is shown in Fig. 8c. As shown, the proposed algorithm accurately represents the shape of the curb, which automatically provides a useful reference border for the driver.

Once the pixels defining the curb are identified, by applying the IPM, one gets their coordinates in the vehicle-centered reference frame $\mathbf{w}_{vis}$. The system has now two sets of coordinates: the ones estimated by the map-based approach and the ones extracted by the ADAS camera feed. If the HUD projection were based on either of the two simply based on the distance, the driver could perceive a discontinuity when the projection switches from $\mathbf{w}_{pos}$ to $\mathbf{w}_{vis}$. A smoothing mechanism avoids this problem. The coordinates that are used for the HUD projections are computed as follows:

$$\mathbf{w} = \lambda(k)\mathbf{w}_{vis} + (1 - \lambda(k))\mathbf{w}_{pos} \quad (5)$$

where

- $\lambda(k)$ is 0 when the distance k from the apex of the corner is higher than 60 m;
- $\lambda(k)$ is 1 when the distance k from the apex of the corner is lower than 30 m;
- $\lambda(k)$ is linearly interpolated between 0 and 1 when the distance from the corner is between 60 and 30 m.

This mechanism yields a smooth transition from one approach to the others. The HUD projection method is then based on $\mathbf{w}$.

IV. RESULTS AND DISCUSSION

The methodology described in the previous section is tested against experimental data. A section of the results is depicted in Fig. 9. The driver approaches (Fig. 9a-9b) and drives through (Fig. 9c-9f) a turn at high speed. From figure, a few comments are due:

- When approaching the turn, the curb localization and projection is mainly based on the map, yet the visual cue is correctly on the actual curb.
- Although the vehicle's attitude changes significantly during this maneuver (in Figure 9b the driver is braking hard), the pitch and roll correction yields an accurate HUD projection.
- When the vehicle gets closer to the turn, the curb is detected through the vision system. Also in this case the projection is precisely on the curb.
- When the vehicle is close to the apex of the corner, the curb projection would require a very large HUD. The frame in the picture represents a reasonable size for a modern automotive HUD. Note that in case of right hand corners, the HUD would need to extend beyond the windscreen.

V. CONCLUDING REMARKS

In this work, we discussed the use of AR HUD in racing applications. The paper details the methodological steps to detect and project significant features. In particular, we focused our framework on the detection and projection of the corner curbs. The application is based on merging the information extracted from a ADAS camera with the map and localization information. Moreover, the precise projection requires to account for the the roll and pitch angles.

The extended experimental validation informs the following considerations:

- in racing applications, due to high-speed, features need to be highlighted when they are several hundred meters away (*e.g.*, 100 – 150 m). This requires the use of features localization techniques beyond computer vision.
- In order to provide actionable information to the driver, the HUD screen needs to have a large FOV (*e.g.*, at least as wide as the windshield and half its height).
- Even using a large HUD, some features in non-straight parts of the circuit may still be outside of the HUD field of view.

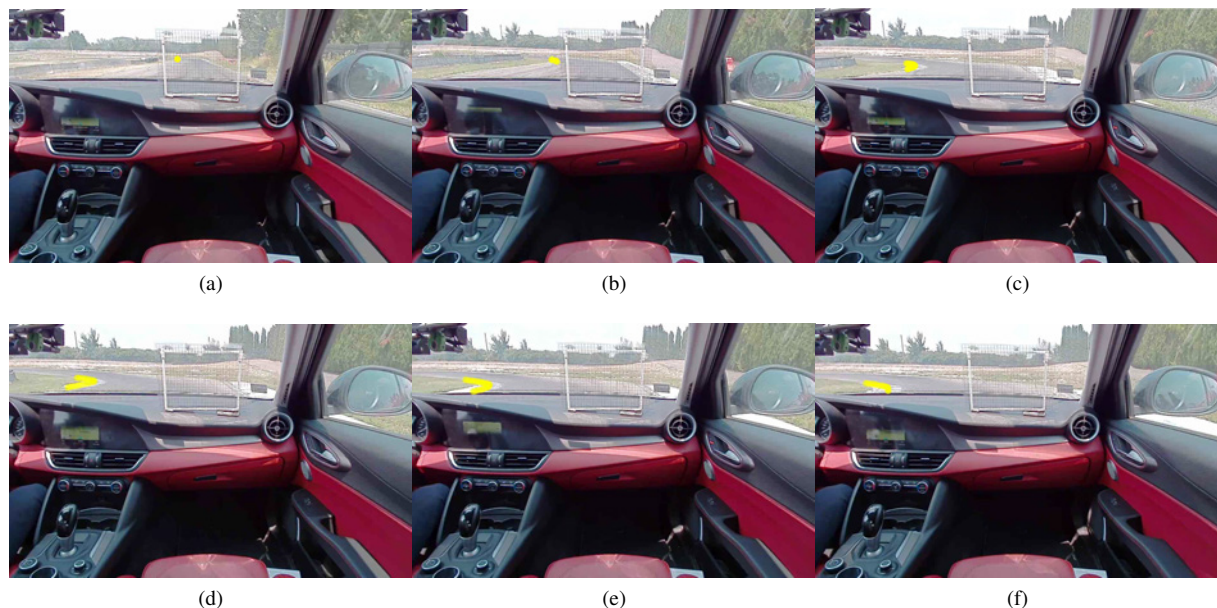


Fig. 9: Simulation of the detected curb's projection on the HUD. As shown in (c)-(f) the feature would exceed the windshield, making this application unfeasible.

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